



OpenPipeline Fundamentals

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Understanding the Unified Data Ingestion Framework

This notebook introduces OpenPipeline, Dynatrace's unified data processing framework for logs, traces, metrics, and events.

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Prerequisites

-  Access to a Dynatrace environment with log data
-  DQL query permissions (viewer role minimum)
-  Basic understanding of log management concepts

1. What is OpenPipeline?

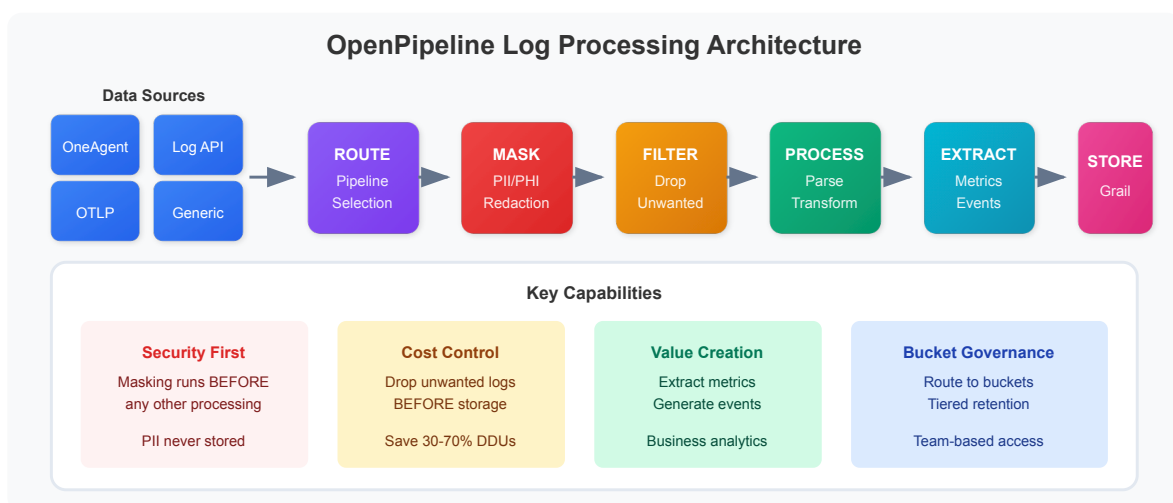
OpenPipeline is Dynatrace's unified data ingestion and processing framework that replaces classic log ingestion. It provides:

- **Unified Processing:** Single framework for logs, metrics, traces, and business events
- **Real-time Transformation:** Parse, enrich, mask, and route data at ingestion
- **Grail Storage:** Direct integration with Dynatrace's data lakehouse
- **Flexible Routing:** Send data to different buckets with custom retention
- **Cost Control:** Drop unnecessary data before storage

OpenPipeline vs Classic Log Ingestion

Feature	Classic Logs	OpenPipeline v2.0
Data Processing	Post-ingestion	At ingestion time
Storage	Log Storage v1	Grail Data Lakehouse
Query Language	Limited	Full DQL Support
Retention	Global	Per-bucket configurable
Data Masking	Limited	Full regex support
Parsing	Basic	DPL (Dynatrace Pattern Language)
Custom Routing	No	Yes, by content/source

2. OpenPipeline Architecture



3. Exploring Your OpenPipeline Data

Let's start by discovering what data sources and pipelines are active in your environment.

```
// Discover data sources feeding OpenPipeline
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// See which pipelines are processing your logs
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.pipelines}
| sort log_count desc
```

```
// Check storage bucket distribution
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.system.bucket}
| sort log_count desc
```

4. Key OpenPipeline Fields

OpenPipeline adds metadata fields to every log record:

Pipeline Metadata

Field	Description	Example
<code>dt.openpipeline.source</code>	How the log was ingested	<code>oneagent</code> , <code>/api/v2/logs/ingest</code> , <code>/api/v2/otlp/v1/logs</code>
<code>dt.openpipeline.pipelines</code>	Pipeline(s) that processed the log	<code>["logs:pipeline_Default_Pipeline_2798"]</code>
<code>dt.system.bucket</code>	Storage bucket name	<code>default_logs</code> , <code>custom_logs</code>

Core Log Fields

Field	Description
timestamp	When the log was generated
content	The log message body
loglevel	Log severity (ERROR, WARN, INFO, DEBUG, NONE)
status	Status string (alternative to loglevel)
log.source	Source identifier (e.g., "Container Output")
log.iostream	Stream type (stdout, stderr)

```
// View a sample log record with all key fields
fetch logs, from: now() - 1h
| fields timestamp, content, loglevel, status, log.source, log.iostream,
      dt.openpipeline.source, dt.openpipeline.pipelines, dt.system.bucket
| limit 5
```

```
// Analyze log levels in your environment
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

5. Data Sources Explained

OneAgent (`oneagent`)

Logs collected automatically by Dynatrace OneAgent from:

- Container stdout/stderr
- Process log files
- System logs

Log Ingest API (`/api/v2/logs/ingest`)

Logs sent directly via the Dynatrace API:

- Custom application logs
- Third-party integrations
- Cloud provider logs (AWS, Azure, GCP)

OTLP (`/api/v2/otlp/v1/logs`)

OpenTelemetry Protocol logs:

- OpenTelemetry Collector
- Fluent Bit with OTLP output
- Custom OTLP exporters

```
// Compare volume by data source
fetch logs, from: now() - 24h
| summarize {
    log_count = count(),
    unique_hosts = countDistinct(dt.entity.host)
}, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// Logs per hour by source (trend analysis)
fetch logs, from: now() - 24h
| makeTimeseries {log_count = count()}, by: {dt.openpipeline.source}, interval:
1h
```

6. Pipeline Stages Overview

OpenPipeline processes data through ordered stages:

Stage 1: Routing

- Matches incoming data to the appropriate pipeline
- Based on source, content, or metadata

Stage 2: Masking (Security)

- Redacts sensitive data BEFORE processing
- Protects PII, credentials, secrets
- Applied early for security compliance

Stage 3: Filtering

- Drops unwanted records
- Reduces storage costs
- Removes noise (debug logs, health checks)

Stage 4: Processing

- Parses structured data from content
- Adds enrichment fields
- Transforms and normalizes data

Stage 5: Extraction

- Creates metrics from log data
- Generates events and business events

Stage 6: Storage

- Routes to appropriate Grail bucket
- Applies retention policies

7. Environment Summary

Let's get a complete picture of your OpenPipeline environment:

```
// Complete environment summary
fetch logs, from: now() - 1h
| summarize {
  total_logs = count(),
  unique_sources = countDistinct(dt.openpipeline.source),
  unique_buckets = countDistinct(dt.system.bucket),
  unique_hosts = countDistinct(dt.entity.host),
  error_count = countIf(loglevel == "ERROR" OR status == "ERROR"),
  warn_count = countIf(loglevel == "WARN" OR status == "WARN")
}
```

```
// Top log sources by entity
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {log_count = count()}, by: {host.name, log.source}
| sort log_count desc
| limit 15
```



Summary

In this notebook, you learned:

- ✓ **What OpenPipeline is** - Dynatrace's unified data processing framework
 - ✓ **Architecture** - Data flow from sources through processing to Grail
 - ✓ **Key fields** - `dt.openpipeline.source` , `dt.openpipeline.pipelines` , `dt.system.bucket`
 - ✓ **Data sources** - OneAgent, Log Ingest API, OTLP
 - ✓ **Pipeline stages** - Routing, Masking, Filtering, Processing, Extraction, Storage
-



Next Steps

Continue to **OPLOGS-02: Migration Guide** to learn how to migrate from classic log ingestion to OpenPipeline v2.0.



References

- [OpenPipeline Documentation](#)
 - [Grail Data Lakehouse](#)
 - [DQL Reference](#)
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